

Development of Controlling Algorithm for DEC-IEC-DX **Hybrid Cooling Systems**



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Gantt Chart

Activity	(7–21 Jan)	(21 Jan–4 Feb)	(4–18 Feb)	(18 Feb–4 Mar)	(4–18 Mar)	(18 Mar–1 Apr)	(1–15 Apr)	15–24 Apr)
Control algorithm refinement from minor project								
Designing control flow for hardware implementation								
Electrical circuit design								
Hardware deployment planning (component identification, procurement)								
External testing of circuit before integration								
Deployment of hardware circuit in actual system								
Experimental readings & system performance analysis								
Validation of new system with control algorithm								
Introduction of additional sensors (water level, water temp, coil temp) & control								
Results compilation, conclusion & presentation preparation								

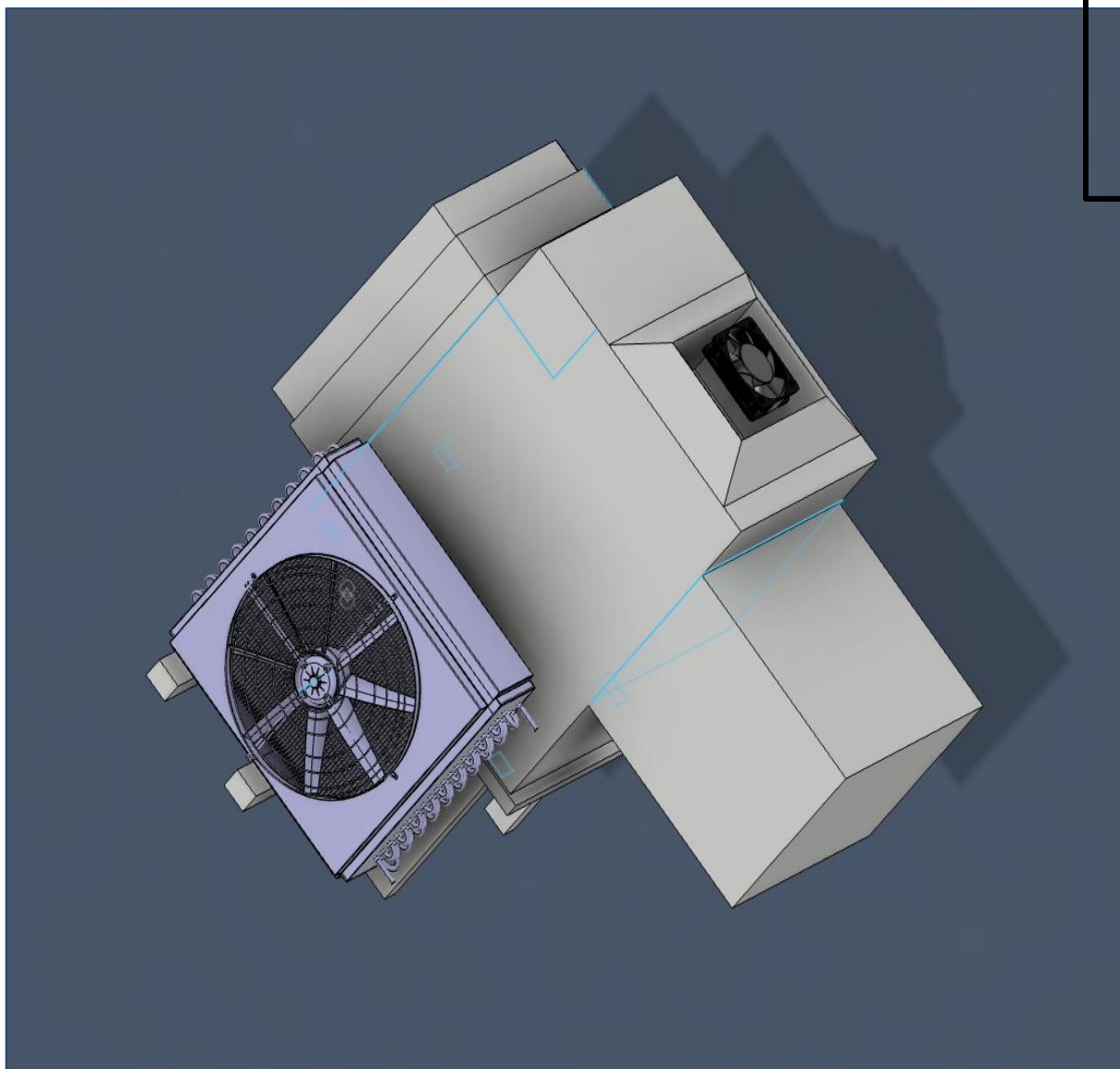
INTRODUCTION

- Cooling systems are essential for maintaining thermal comfort and energy efficiency in modern buildings
- Conventional **DX (Direct Expansion)** systems provide strong cooling but consume high energy
- **Evaporative cooling (IEC & DEC)** offers energy-efficient alternatives, especially in hot climates
- However, performance of these systems varies significantly with **ambient conditions and airflow**
- There is a need for an **intelligent control strategy** to select the most efficient cooling mode dynamically
- This project focuses on developing a **smart control algorithm** integrating IEC, DEC, and DX systems
- The system aims to **optimize COP based on the ambient condition and improve cooling effectiveness using smart controller.**

Motivation behind the project

- Lack of Proper Controlling Algorithms: Current systems often do not have sophisticated or adaptive control strategies to coordinate the interaction between indirect evaporative cooling and DX refrigeration.
- No Reliable Tool for System Sizing: Designers lack effective simulation or sizing tools specific to these hybrid systems, making optimization and customization difficult.
- Limited Automation for Energy-Efficient Mode Switching: Most systems do not incorporate automated mode switching based on minimum power consumption or maximum COP, resulting in inefficient operation and higher energy usage

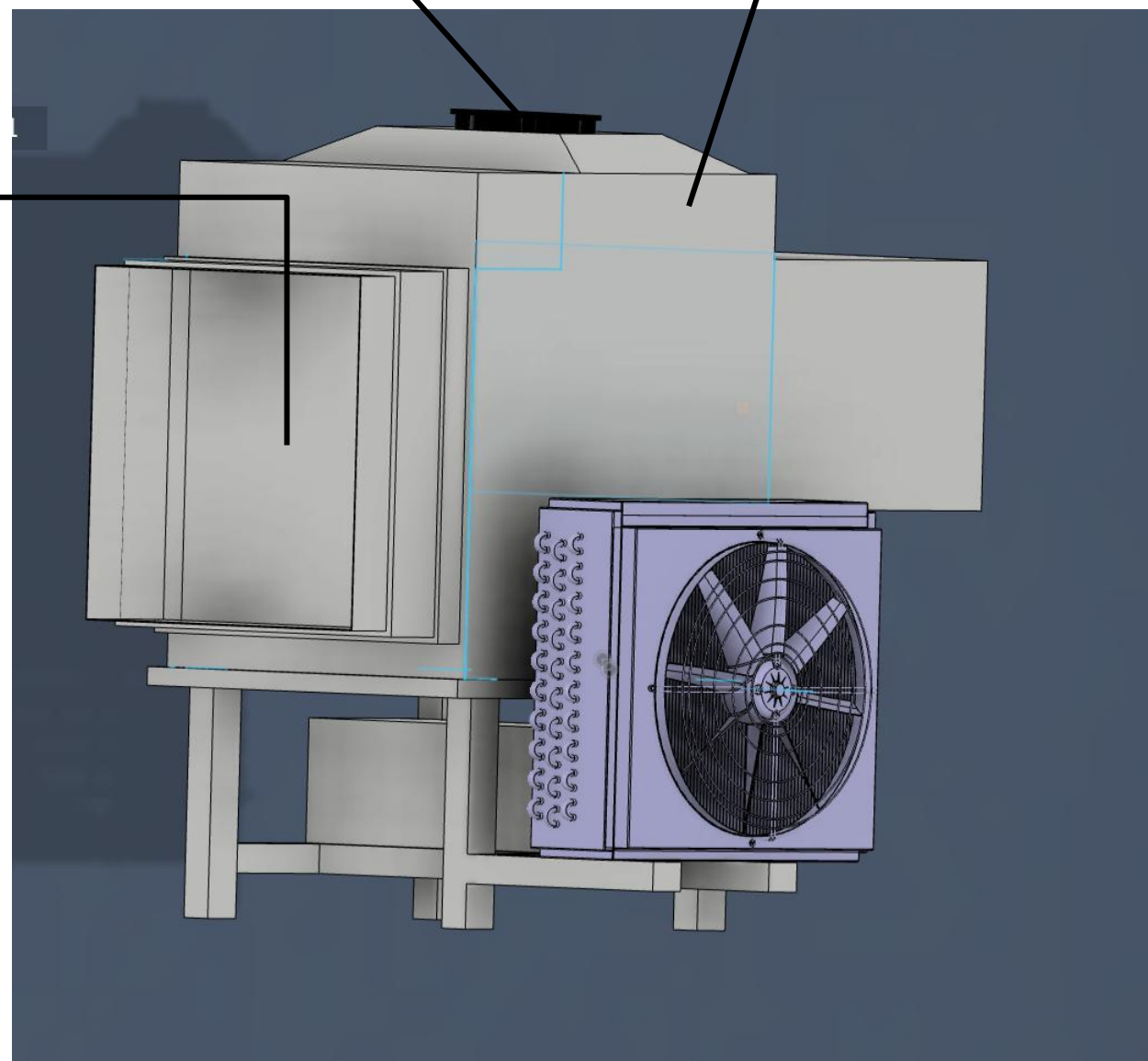
CAD Model of Hybrid Sytem



INLET

EXHAUST

IEC & DEC



Literature Review on Air Flow and Water Flow Velocity

Research Title	Journal and year published	Author	Focus	Key Findings
Experimental investigation of evaporative cooling systems: influence of operating parameters on performance	Energy and Buildings (2020)	Granata et al.		Optimal water flow rate 4.76 L/min·m²
Impact of Airflow on Evaporative Cooling System by Analysis of Cooling Efficiency	SciEn Conference Series: Engineering (2025)	Rahman et al.		Cooling efficiency improves at lower airflow

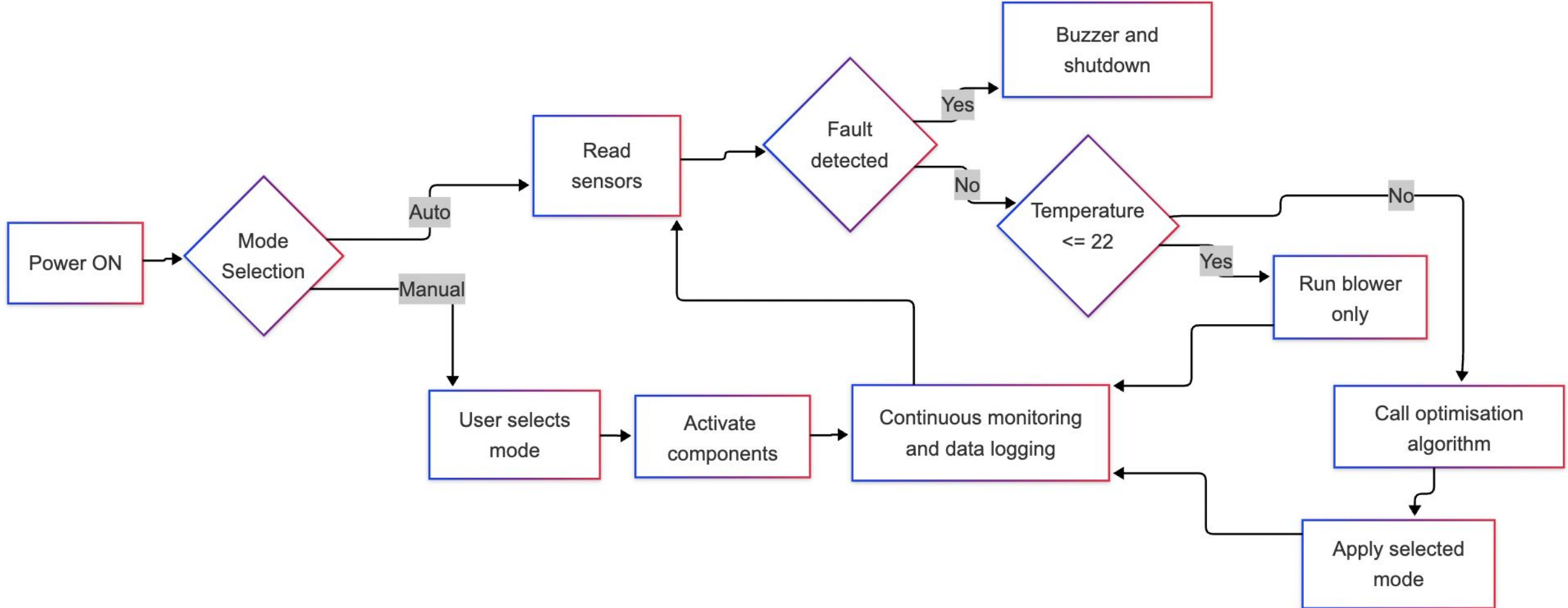
Literature Review on DEC &IEC

Research Title	Journal and year published	Author	Focus	Key Findings
A Passive Evaporative Cooling System by Natural Ventilation	Building and Environment (1996)	Giabaklou & Ballinger	Passive evaporative cooling	Improved indoor comfort in hot climates
An Evaporative and Desiccant Cooling System for Air Conditioning in Humid Climates	Journal of the Brazilian Society of Mechanical Sciences and Engineering (2005)	Camargo et al.	Air velocity impact	Lower air velocity → higher cooling efficiency
Mathematical model for heat and mass transfer in evaporative cooling equipment	International Journal of Heat and Mass Transfer (1998)	Alonso et al.	Heat transfer modeling	Developed heat exchanger analysis model

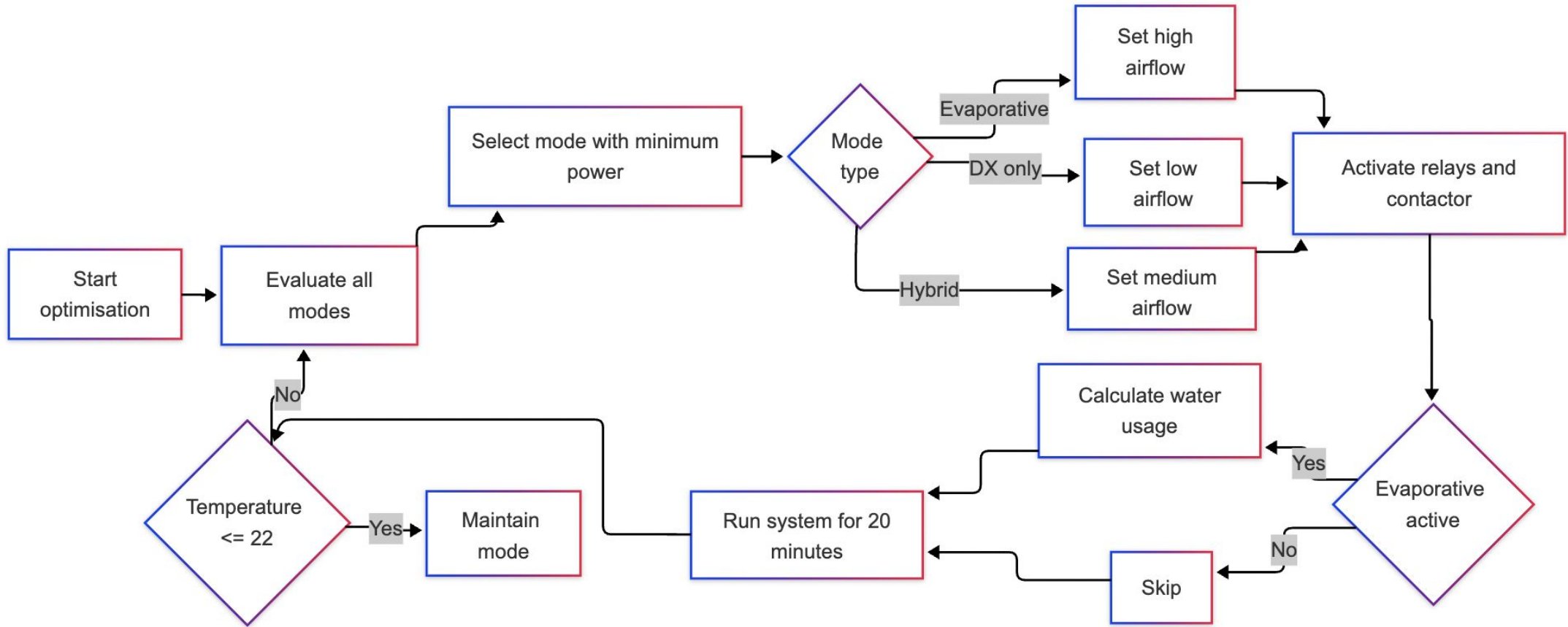
COMPONENTS

S.No	Component	Description
1	Arduino Nano ESP32 Microcontroller	Main controller executing the cooling control algorithm
2	SHT41 Temperature and Humidity Sensor	Measures ambient temperature and relative humidity
3	DS18B20 Temperature Sensors	Used for inlet and outlet air temperature measurement
4	PZEM-004T Energy Meter Module	Measures voltage, current, power, and energy consumption
5	Opto-isolated Relay Modules	Controls pumps, fans, and electrical loads
6	Solid State Relay (SSR-40DA)	High power switching for AC loads
7	AC Contactor	Used for switching the DX compressor
8	OLED Display	Displays system status and sensor readings
9	Micro SD Card Data Logger	Stores experimental data for analysis
10	Water Level and Vibration Sensors	Safety monitoring for pump dry run and equipment vibration

Control Flowchart



Control Flowchart



Parameters	IEC+DEC+DX (450)	DX (450)	IEC+DEC+DX (600)	DX (600)	IEC+DEC+DX (750)	DX (750)
Ambient Temperature, T (°C)	33	33	33	33	33	33
Ambient Relative Humidity (%)	38	38	38	38	38	38
Wet Bulb Temperature, Tw (°C)	21.79	21.79	21.79	21.79	21.79	21.79
Outlet Temperature, T (°C)	15.6	17.1	16.6	19.2	17.6	23.1
Outlet Relative Humidity (%)	92	60	93	70.2	92	58.9
Airflow (CFM)	450	450	600	600	750	750
Mass Flow Rate (kg/s)	0.18	0.18	0.25	0.25	0.31	0.31
Temperature Change, ΔT (°C)	17.4	15.9	16.4	13.8	15.4	9.9
Enthalpy Change, Δh (kJ/kg)	22.67	28.84	19.49	20.13	16.86	14.54
Power Consumption (kW)	0.93	1.70	0.96	1.82	1.20	1.95
COP	4.04	3.10	5.07	2.76	4.35	2.31
Cooling Effect (kW)	4.00	5.20	4.88	5.04	5.22	4.50
Coil Set Temperature (°C)	—	18	—	18	—	18

Parameters	IEC+DEC+DX (450)	DX (450)	IEC+DEC+DX (600)	DX (600)	IEC+DEC+DX (750)	DX (750)
Ambient Temperature, T (°C)	39	39	39	39	39	39
Ambient Relative Humidity (%)	22	22	22	22	22	22
Wet Bulb Temperature, Tw (°C)	22.02	22.02	22.02	22.02	22.02	22.02
Outlet Temperature (°C)	16.5	24	17	25.8	18.5	28
Outlet Relative Humidity (%)	75	33.4	72	29.4	68	27.7
Airflow (CFM)	450	450	600	600	750	750
Mass Flow Rate (kg/s)	0.18	0.18	0.25	0.25	0.31	0.31
Temperature Change ΔT (°C)	22.5	15	22	13.2	20.5	11
Enthalpy Change Δh (kJ/kg)	25.29	24.5	24.97	23	22.55	19.59
Power Consumption (kW)	1.21	1.91	1.43	2.01	1.20	2.04
COP	5.44	3.34	5.94	3.90	6.60	4.13
Cooling Effect (kW)	6.58	6.37	8.49	7.82	9.70	8.43
Coil Set Temperature (°C)	–	18	–	18	–	18

Conclusion

The proposed hybrid system gives better and lower outlet temperatures than DX alone. This suggests that cooling effect is better in hybrid system as compared with DX. Higher Compressor power in DX leads to lower COP while energy efficient cooling in Hybrid system leads to higher COP. Our system monitors ambient conditions over a fixed period of time i.e collects real time data for computing cooling load. Also, system takes data of current, voltage in order to get glimpse into power consumption. Both of this help to choose mode/combination whose COP is maximum.

Sensors such as ultrasonic sensor and vibration sensor ensure preventing dry running of IEC and DEC pumps and checking of functioning of compressor respectively i.e safety of components is also taken into consideration. This system is industry ready and will provide significant energy savings to the organisation contributing to reduced operation expenses.

Limitations

1. Our automation was deployed on smaller scale i.e relatively lower cooling load. When this system would be implemented on larger scale, inclusion of PLC and other electronic components will increase complexity of the system, suggesting that current system is not suitable for larger cooling loads.
2. For hot and humid conditions, Evaporative cooling modes are not suitable. Hence, inclusion of DX partially is required leading to increased energy consumption.

Future Scope

Large datasets of various parameters such as temperature, humidity, power, mode/combo etc would be stored in the SD card. Anticipation of COP with help of this big data, in advance, will be more accurate and will help in prevention of any errors or malfunctions that can arrive in the system i.e predictive maintenance.